

Detecting Fish Hunger Levels by Monitoring and Classifying Their Underwater Swimming Patterns

Research Proposal

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1. Introduction and Background

Fish farming is one of the most important sources of food and income in Bangladesh, especially for small and mid-sized pond owners in rural areas. It supports millions of livelihoods and plays a direct role in the country's food security. Like most forms of intensive farming, though, its profitability comes down to managing a few key inputs well — and feed is by far the biggest one.

Feed alone can account for around 60% of total production cost in Bangladeshi aquaculture. Most farmers still feed their fish on a fixed schedule, or simply judge by standing at the edge of the pond and watching the surface. This works, but it isn't very precise. It doesn't account for whether fish are actually hungry on a given day, and it can't adapt to day-to-day changes like temperature or how much they were fed the previous round. The result is usually overfeeding: leftover feed that sinks, rots, and slowly drags down water quality, on top of being money that didn't need to be spent.

What we noticed is that fish don't hide their hunger very well — they swim differently depending on how hungry they are. Hungry fish tend to move faster, turn more often, rise toward the surface or feeding zone, and cluster more tightly together. These are all things a camera can actually see. Our proposal is to build a low-cost, camera-based system that watches this behavior underwater and uses a machine learning model to classify the fish's hunger level as **Low**, **Moderate**, or **High** — giving farmers a simple, real-time answer to the question "should I feed right now?" without needing expensive sensors or guesswork.

We want to be upfront that a camera obviously can't measure hunger directly — it's a physiological state, not something visible. What we're actually testing is whether measurable swimming behavior is a reliable enough proxy for hunger to be useful in practice, and whether feeding guided by that behavior beats the fixed-schedule approach most farms use today.

2. Problem Statement

2.1 Inaccurate Feeding Methods

Fixed-time feeding schedules don't check whether fish are actually hungry before the next ration goes in, and judging appetite by eye from the pond's surface is subjective and inconsistent from one farmer to the next. Neither approach gives a real, moment-to-moment signal of feeding demand.

2.2 Wasted Money and Water Quality

Because of this, overfeeding is common. Feed that isn't eaten doesn't just disappear — it settles, decomposes, and can lower dissolved oxygen and raise ammonia in the pond, which in turn can stress the fish and further suppress their appetite. It's a cost problem and an environmental problem at the same time.

2.3 High-Cost Technology Is Out of Reach

Commercial smart-feeding systems do exist, but they're generally built and priced for large, industrial-scale operations — specialized sensors, automated feeders, cloud platforms, and reliable high-bandwidth internet. Small and mid-sized Bangladeshi farms, which make up a large share of

the sector, typically can't justify that cost or infrastructure.

2.4 No Local Tools or Data to Build On

There is currently no simple, low-cost way to estimate hunger level for locally farmed fish species, and just as importantly, no public dataset connecting swimming behavior to hunger state for these species and pond conditions. Models built on other datasets (different species, clearer water, lab tanks) don't necessarily transfer well to a turbid Bangladeshi pond. That gap is part of what this project is trying to fill.

3. Research Objectives

- **Objective 1:** Build a low-cost, camera-based setup to record underwater fish behavior under realistic pond conditions.
- **Objective 2:** Track fish movement and extract behavioral features — swimming speed, turning frequency, vertical position, and schooling density.
- **Objective 3:** Train a machine learning / deep learning model (LSTM or CNN-LSTM) to classify hunger level into Low, Moderate, or High from these behavioral features.
- **Objective 4:** Evaluate whether need-based feeding, guided by the model's output, is more efficient than conventional fixed-schedule feeding.

4. Research Questions and Contribution

Research Questions

- **RQ1.** Can underwater swimming patterns reliably indicate a fish's hunger level?
- **RQ2.** Can a low-cost, camera-only computer vision system meaningfully help farmers decide when to feed?

Our Contribution

- A labeled dataset linking swimming behavior to hunger state for locally farmed fish — something that is currently scarce or non-existent.
- A simple, real-time hunger indicator designed with small and mid-sized farms in mind, not industrial-scale operations.
- A clearer picture of which behavioral cue — speed, turning, depth, or schooling — carries the strongest signal of hunger, which could be useful for behavioral research beyond feeding alone.

5. Proposed Methodology

The system will follow a straightforward pipeline, moving from raw video to a usable feeding signal:

- **Record.** Film fish underwater at different times since their last feeding, so that each video clip has a known ground-truth hunger state to train and test against.
- **Track.** Detect the fish in each frame and follow their paths across frames to reconstruct movement trajectories.
- **Extract features.** From those trajectories, compute the four behavioral indicators — swimming speed, turning frequency, vertical position, and schooling density — over short time windows.
- **Train.** Feed these behavioral sequences into an LSTM or CNN-LSTM model, which learns to classify them into Low, Moderate, or High hunger. Simpler baselines (e.g. logistic regression, random forest) will also be tested, so we can tell whether the more complex temporal model is actually earning its keep.
- **Test.** Evaluate the trained model on new video it hasn't seen before, and run a comparison between need-based feeding (guided by the model) and fixed-schedule feeding, looking at feed used, fish growth, and water quality.

The methodology deliberately favors components that are realistic for a small farm: a single waterproof camera rather than a bank of sensors, a model light enough to run on modest hardware, and features that can be computed from ordinary underwater video rather than specialized lab equipment.

Underwater Video → Fish Detection & Tracking → Feature Extraction → LSTM / CNN-LSTM → Hunger Level (Low / Moderate / High) → Feeding Decision

6. Expected Outcomes

- A working prototype that classifies fish hunger level from underwater video with reasonable accuracy.
- A labeled dataset connecting swimming behavior to hunger state — currently scarce for locally farmed species.
- A ranking of which movement feature (speed, turning, depth, or schooling) is the strongest predictor of hunger.
- Evidence on whether need-based feeding reduces wasted feed and supports better water-quality outcomes compared with fixed-schedule feeding.

We see this less as a finished product and more as a proof that the idea holds up — a working prototype and an honest evaluation of its accuracy and impact, which can be built on further after this proposal stage.

7. References

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